Impeller's hydraulic radial thrust calculation

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1. Purpose

This document defines the guidelines for the calculation of the hydraulic radial thrust acting on an impeller in centrifugal pumps.

The procedure is applicable to the plant of Nova Milanese and the plant of La Spezia.

2. Abbreviations

BEP Best Efficiency Point

3. Scope

This procedure applies to all the impellers in centrifugal pumps to estimate the hydraulic radial thrust during the operation. The calculation approaches are based on *J. F. Gulich's Centrifugal Pumps* and on *ANSI/HI-1.3*. As general recommendation, Gulich method shall be used in high-energy applications and when ANSI/HI-1.3 method is not applicable, while ANSI/HI-1.3 method shall be considered as reference for not high-energy pumps, within its applicability conditions

4. Calculation method - J.F. Gulich's Centrifugal Pumps

The calculation approach considered in the following is based on *Table 9.7* of *J. F. Gulich's Centrifugal Pumps*. It is recommended for all the type of pumps.

The radial thrust is calculated based on the following formula:

$$F_R = k_R \cdot \rho \cdot g \cdot H \cdot d_2 \cdot b_{2tot}$$

Where:

- **b_{2tot} [m]** is the impeller outlet width including shrouds.
- **d_2 [m]** is the impeller diameter.
- **F_R [N]** is the impeller radial thrust.
- **g [m/s^2]** is the acceleration of gravity.
- **H [m]** is the impeller head.
- **k_R [-]** is the radial thrust coefficient.
- **ρ [kg/m^3]** is the fluid density.

The radial thrust coefficient **k_R** is dependent on the ratio q^* between the operating flowrate and the best efficiency flow rate. Different formulas are defined for force type (i.e. steady or unsteady) and for the casing type (i.e. single volute, double volute, twin volute, annular casing and diffuser).

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4.1. Steady radial force

The calculation of the steady radial force is performed based on the following approach.

4.1.1. Single volute

In case of single volute, the radial thrust coefficient is calculated as in the following:

a) Operating condition $q^* = 0$:

k_{R0} is calculated from *Figure 4.1* based on total specific speed:

$$n_q = \frac{n \cdot \sqrt{Q_{BEP}/n_{eye}}}{H_{BEP}^{0.75}}$$

$$n_{q,tot} = n_q \cdot n_{eye}$$

Where:

- **n [rpm]** is the rotational speed.
- **n_{eye} [-]** is the number of impeller entries (1 or 2).
- **n_q** and **$n_{q,tot}$** are the actual and the total specific speeds.
- **Q_{BEP} [m³/s]** and **H_{BEP} [m]** are the flowrate and the head at BEP.
- *Equation 3.12* is

$$\frac{\Delta d}{d_{sp}} = 0.004 \cdot \left(\frac{100}{d_{sp}}\right)^{0.53}$$

Where **Δd [mm]** is the diametral clearance, **d_{sp} [mm]** is the wear ring diameter.

Curves 1 and 2 are applicable for wear ring's clearance as per *J. F. Gulich's Centrifugal Pumps. Actually*, different clearances are usually adopted. As a precaution, Curve 2 is used in the calculation.

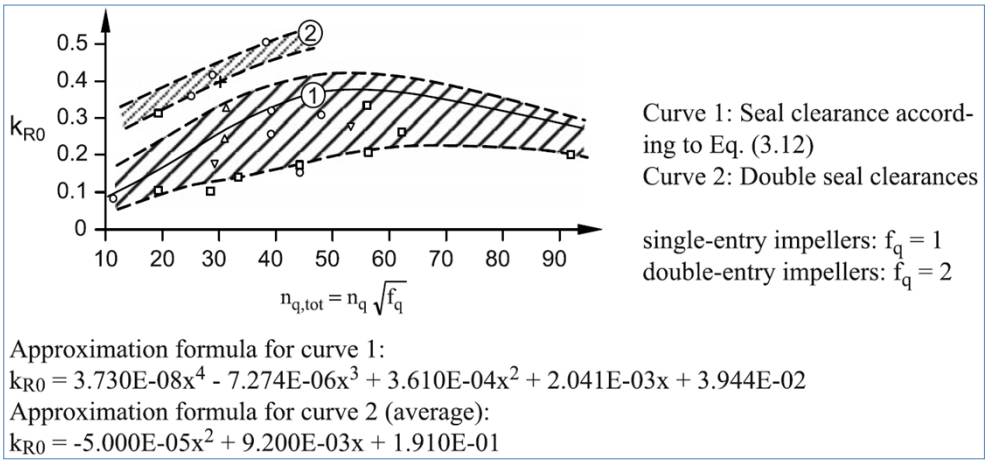

Figure 4.1 Static radial force coefficients for single volutes at shut-off.

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b) Operating condition $q^* = 0.5$:

k_{R50} is calculated based on the following formula (refer to *Figure 4.2*).

$$k_{R50} = -2.515 \cdot 10^{-6} \cdot n_{q,tot}^{3} + 2.633 \cdot 10^{-4} \cdot n_{q,tot}^{2} - 2.197 \cdot 10^{-3} \cdot n_{q,tot} + 3.429 \cdot 10^{-2}$$

c) Operating condition $0 < q^* < 1$:

k_R is calculated from *Figure 4.1* based on the following formula (refer to *Figure 4.2*).

$$k_R = \left(k_{R0} - k_{R,opt}\right) \cdot \left(1 - q^{*2}\right) + k_{R,opt}$$

Where:

- **k_{R0}** is the radial thrust factor at shut-off defined in the previous point for $q^*=0$.
- **$k_{R,opt}$** is the radial thrust factor at shut-off defined in the previous point for $q^*=1$.

d) Operating condition $q^* = 1$:

k_{R100} can be calculated based on the following formula (refer to *Figure 4.2*) or selected within the range 0.03 – 0.08.

$$k_{R100} = 4.562 \cdot 10^{-5} \cdot n_{q,tot}^{2} - 1.894 \cdot 10^{-3} \cdot n_{q,tot} + 3.179 \cdot 10^{-2}$$

As general consideration, it is recommended to check the sensitivity of this parameter within its range.

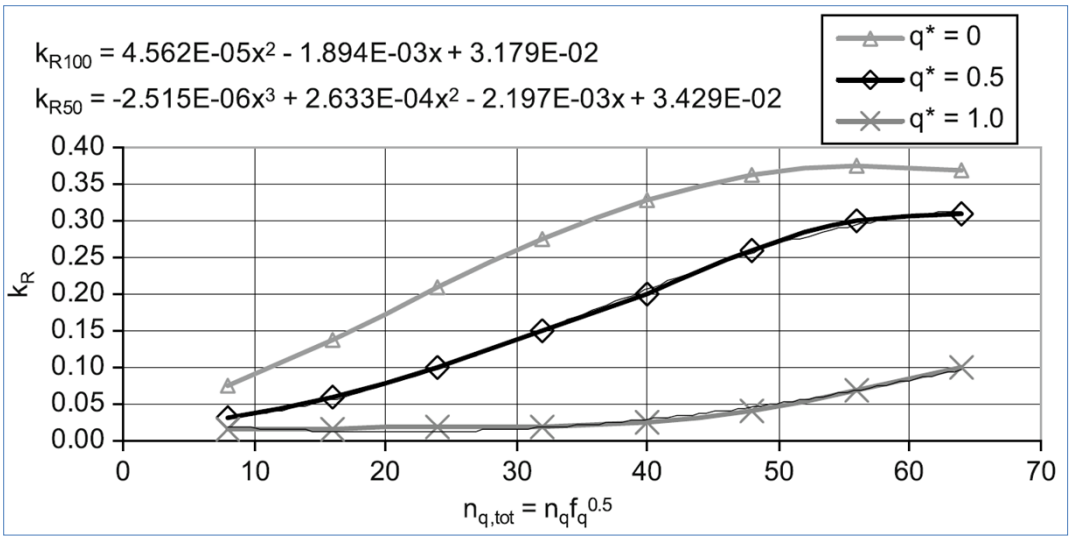

Figure 4.2 Static radial force coefficients for single volutes.

e) Operating condition $q^* > 1$:

k_R is calculated from *Figure 4.1* based on the following formula.

$$k_R = 0.09 \cdot q^*$$

It is possible to use $k_{R,opt}$ instead of factor 0.09.

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4.1.2. Double volute

In case of double volute, the radial thrust coefficient is calculated as in the following:

$$k_{R,Dsp} = F_{Dsp} \cdot k_{R0}$$
$$F_{Dsp} = 1.75 - 0.0083 \cdot \varepsilon_{sp}$$

Where:

- **F_{Dsp} [-]** is a factor calculated based on *Figure 4.3*.
- **kR0 [-]** is the radial thrust coefficient calculated in Section 4.1.1 for $q^*=0$.
- **ε_{sp} [°]** is the wrap angle of inner volute.

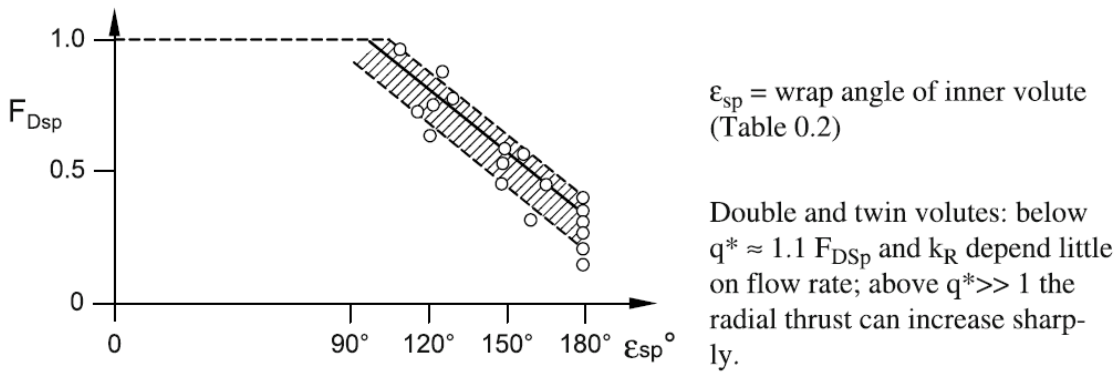

Figure 4.3 Factors for radial force coefficients of double volutes.

Below $q^* \approx 1.1$, F_{Dsp} and k_R depend little on flow rate; above q*>> 1, the radial thrust can increase sharply. If wrap angle ε_{sp} is between 90° and 180°, the radial force coefficient for q* = 0.5 or q* = 1.0 can be estimated from the following interpolation formula:

$$f = \frac{k_{Rx}}{k_{R0,SV}} + \left(1 - \frac{k_{Rx}}{k_{R0,SV}}\right) \cdot \frac{\varepsilon_{sp} - 90}{90}$$

$$k_R = k_{R0,SV} \cdot F_{Dsp} \cdot f$$

Where:

- **k_{Rx} [-]** is the single volute's radial thrust coefficient calculated in Section 4.1.1 for $q^*=0.5$ or 1, on the basis of which point is being calculated.
- **$k_{R0,SV}$ [-]** is the single volute's radial thrust coefficient calculated in Section 4.1.1 for $q^*=0$.

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4.1.3. Twin volute

In case of twin volute, the radial thrust coefficient is calculated as in the following:

$$k_{R,Zsp} = (0.3\ to\ 0.5) \cdot k_{R,Dsp}$$

Where:

- **$k_{R,Dsp}$ [-]** is the double volute's radial thrust coefficient calculated in Section 4.1.2.

4.1.4. Annular casing

In case of annular casing, the radial thrust coefficient is calculated as in the following:

a) Operating condition $q^* = 0$:

$$k_{R0} = 0.03\ to\ 0.1$$

b) Operating condition $q^* = 1$:

$$k_{R,opt} = 0.1\ to\ 0.2$$

c) Other operating conditions:

$$k_R = k_{R0} \cdot \left(1 + q^* + a \cdot q^{*2}\right)$$

Where:

- **k_{R0} [-]** is the annular casing's radial thrust coefficient previously calculated for $q^*=0$.
- **o** is a coefficient depending on geometry. Tentatively can be considered equal to 0.18.

4.1.5. Diffuser

In case of diffuser, the radial thrust coefficients for $q^*=0$ and $q^*=1$ are defined as in the following:

$$k_{R0} = 0.03\ to\ 0.1$$

$$k_{R,opt} = 0.02\ to\ 0.09$$

4.2. Unsteady radial force

A similar approach as per *4.1* is adopted for unsteady radial force's calculation:

a) All type of volutes and annular casings, except single-channel impellers:

$$k_{R,dyn} = 0.07\ to\ 0.12\ \ for\ \ q^* < 0.5$$

$$k_{R,dyn} = 0.01\ to\ 0.05\ \ for\ \ q^* = 1$$

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b) Diffusers:

$$k_{R,dyn} = 0.04\ to\ 0.16\ \ for\ \ q^* < 0.5$$

$$k_{R,dyn} = 0.01\ to\ 0.09\ \ for\ \ q^* = 1$$

5. ANSI/HI-1.3 method

ANSI/HI-1.3 allows to estimate the radial thrust for the impellers in centrifugal during the operation in single or double volute casing. In case that diffusers are used, net radial thrust exists, but its magnitude is commonly considered minimized due to the equally spaced passages and is considered negligible. This method applies to circular (concentric) volute as well, which are commonly used on pump design with specific speeds of 20 or less

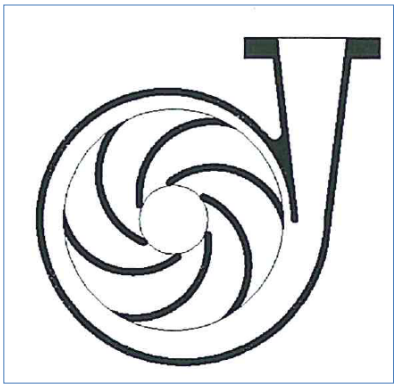

Figure 5.1 Single volute.

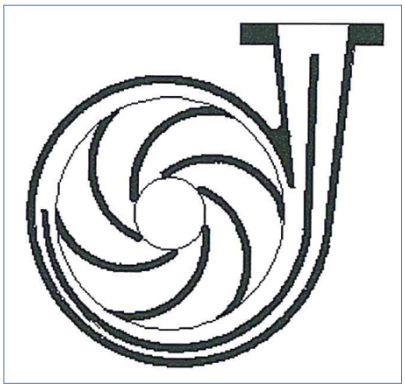

Figure 5.2 Double volute.

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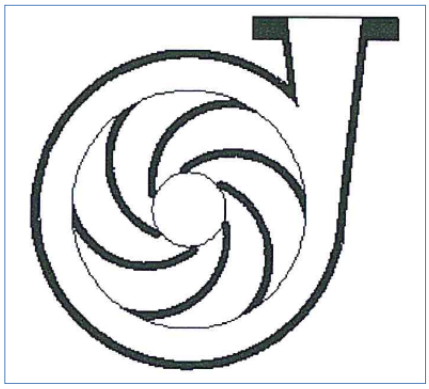

Figure 5.3 Circular (concentric) casing.

Single and the dual volute pumps are commonly designed to uniform mean velocity around the volute at the BEP flow. For this reason, the radial thrust is not usually significant and close to zero at BEP. For operation at flow rates lower or higher than BEP, the pressure distribution is not uniform, resulting in a radial thrust on the impeller. The magnitude and the direction of the radial thrust change based on the flow rate.

This method is considered reliable if applied for impeller within the specific speeds 9.7 – 67 for single and double volute pumps and for specific speed 20 for concentric volute.

The impeller's radial thrust is calculated based on the section *1.3.5* of *ANSI/HI-1.3 Rotordynamic Centrifugal Pumps*:

$$F_R = K_R \cdot H \cdot \rho \cdot g \cdot D_2 \cdot b_2$$

Where:

- **K_R** is the thrust factor, which varies with rate of flow and specific speed. Refer to *Figure 5.5* for single volute and to Figure 5.6 for double volute.
- **H [m]** is the head developed by the single stage.
- **ρ [kg/m³]** is the specific gravity of the operating fluid.
- **g [m/s²]** is the acceleration of gravity.
- **D_2 [m]** is the impeller diameter.
- **b_2 [m]** is the impeller width at discharge including shrouds.

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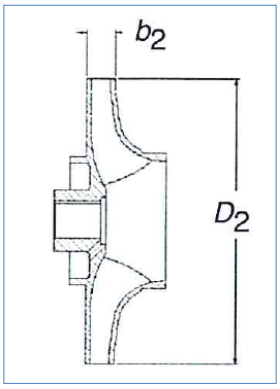

Figure 5.4 Impeller's reference sketch.

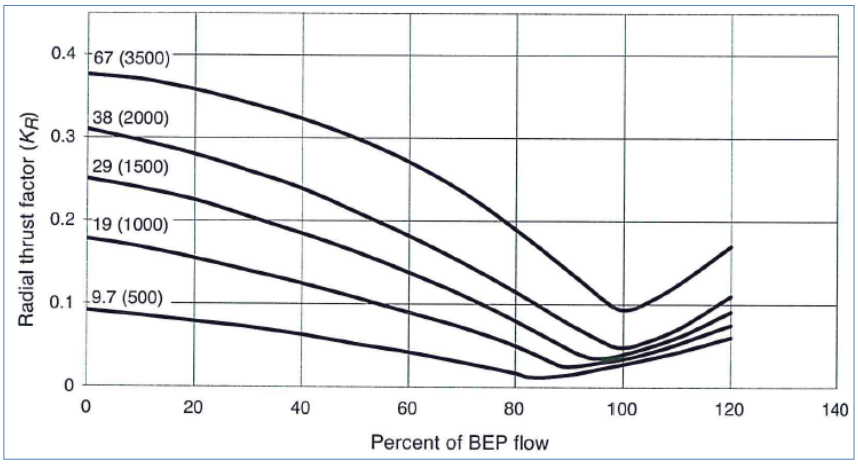

Figure 5.5 Radial thrust factor for single volute with various specific speed.

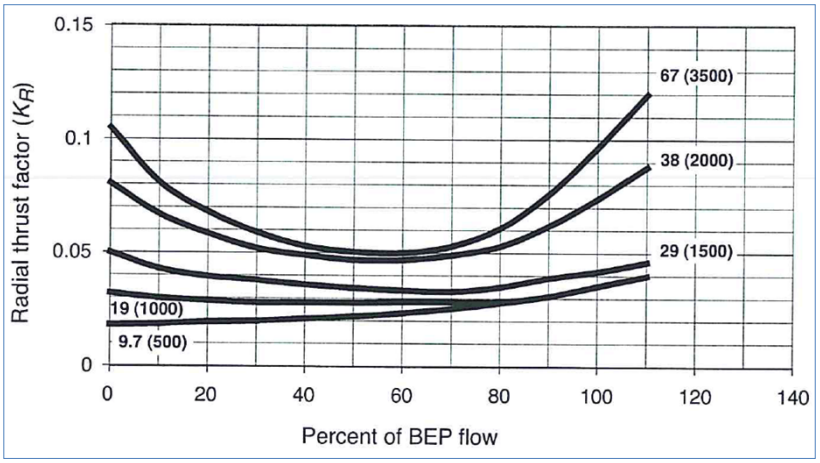

Figure 5.6 Radial thrust factor for double volute with various specific speed.

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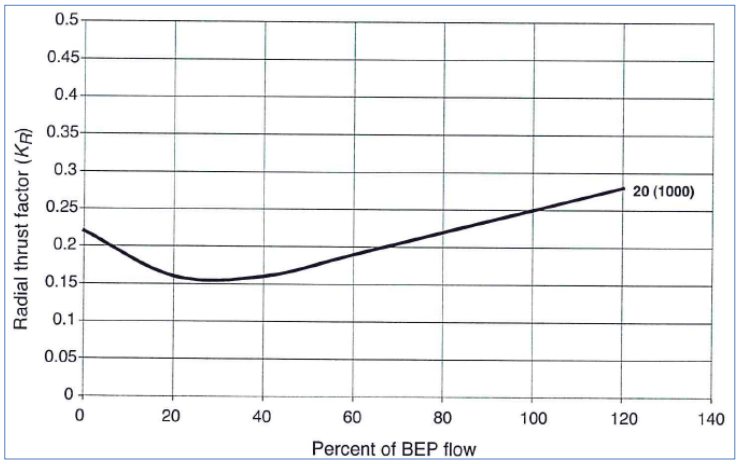

Figure 5.7 Radial thrust factor for circular (concentric) casing typical for specific speed 20 (1000).

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6. Radial thrust direction

For single volute pumps, the highest value of radial thrust is typical at zero flow in a direction from the shaft centreline towards the narrower part of the volute casing. As the flowrate increases, the direction shifts towards the wider volute area. In general, typical direction of the radial thrust is reported in the *Figure 6.1*. These considerations about the direction of each impeller's radial thrust shall be duly considered in the calculation of the rotor's total one.

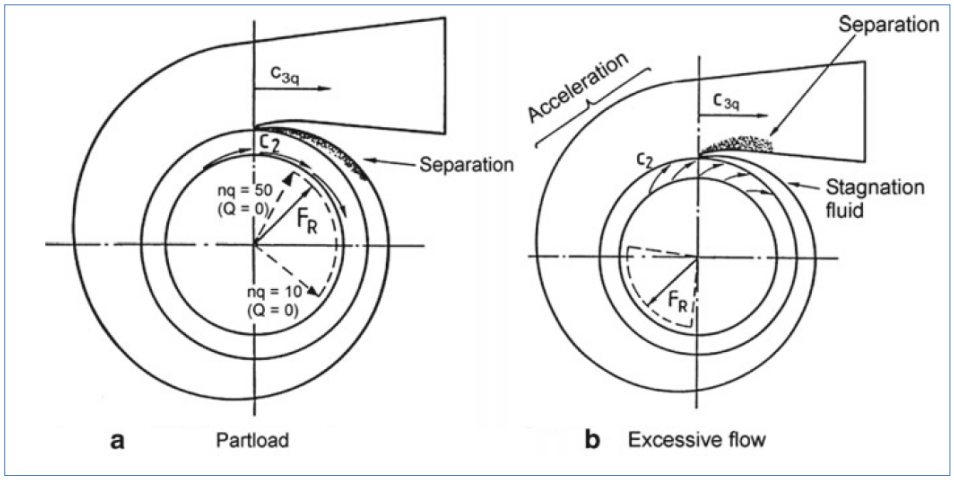

Figure 6.1 Radial thrust direction.

7. References

Rotordynamic Centrifugal Pumps, ASNI/HI 1.3, 2013

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